Computational Complexity

Reading guide and Homework for **Week 08**

1 Reading Guide

This week:

- We learned that the class of functions computed by polynomial-size circuits is the same as the class of functions computable in polynomial time with polynomially-large advice, namely P/poly. §6.3
- We saw that P is exactly the class of functions computed by polytime-generated families of polynomial-size circuits. §6.2
- We saw the lower-bounds on formula size by Nechiporuk (Indexing of OR), Subbotovskaya (parity), and Andreev (indexing of parity). This is not in the book but notes will be provided.

2 Homework

Skill: *Reason about circuits and formulas.*

1. The Boolean matrix product is an operation on two square Boolean matrices $A, B \in \{0,1\}^{n \times n}$, which outputs another square Boolean matrix $C = A \cdot B$ given by:

   $$C_{ij} = \vee_{k=1}^{n} A_{ik} \wedge B_{kj}.$$

   Show that there exists a Boolean formula of size $O(n^3)$ and depth $O(\log n)$ for computing $A \cdot B$ when given A and B.

2. The iterated Boolean matrix product is given as input $n \times n$ Boolean matrices $A_1, \ldots, A_m$, and outputs the product of all of them $A_1 \cdot \cdots \cdot A_m$. Show that there exists a Boolean *circuit* of size $O(n^4)$ and depth $O((\log n)(\log m)$ for computing the iterated matrix product. Why does your construction *not* give a *formula* of size polynomial in n?

3. Let A be the adjacency matrix of a certain graph on n vertices, where we set the diagonal entries to 1 (so we pretend that every vertex as an edge to itself). Let A^n be the iterated product of A with itself n times (which can be computed by a circuit of size n^4 and depth $(\log n)^2$. Show that $(A^n)_{ij} = 1$ iff there exists a path between nodes i and j in G.

   Noticing that path connectivity is NL-hard, show that in fact every problem in NL has a circuit of size polynomial in n and depth $O((\log n)^2)$.

4. Show that a given circuit of size s and depth d can be evaluated in $O(\log s + \log d)$ space.

5. Put it all together to show the following. Let NC_i be the class of problems computable by circuits of polynomial size and $O((\log n)^i)$ depth. Recall that, given a class C, $C/poly$ is the class of problems F for which there exists a problem $F' \in C$ and a polynomially-bounded advice function a such that $F(x) = 1 \iff F'(x, a(|x|)) = 1$. Then show that:

   $$\mathsf{NC}_1 \subseteq \mathsf{L}/poly \subseteq \mathsf{NL}/poly \subseteq \mathsf{NC}_2.$$

Skill: *Understand quantifiers and prove simple lower-bounds in circuit complexity.*

1. Show that there exists a function F in $\mathsf{ESPACE} = \mathsf{DSPACE}(2^{O(n)})$ which cannot be computed by circuits of size $\frac{2^n}{n^2}$. (Hint: the algorithm finds the lexicographically-first function which cannot be computed by circuits of such size, and outputs whatever that function outputs.)

2. Let $\mathsf{E}\Sigma_3$ be the class of decision problems $F : \{0,1\}^* \to \{0,1\}$ for which there exists a deterministic Turing machine M running in time $2^{O(n)}$, such that

$$F(x) = 1 \iff \exists y \forall z \exists w M(x, y, z, w) = 1.$$

   Show that $\mathsf{E}\Sigma_3$ is contained in ESPACE. Show that there exists a function F in $\mathsf{E}\Sigma_3$ which cannot be computed by circuits of size $\frac{2^n}{n^2}$. (*Hint:* same as above; express it as a $\exists\forall\exists$ predicate.)

3. Recall that Σ_3 is the class of decision problems $F : \{0,1\}^* \to \{0,1\}$ for which there exists a polynomial-time machine M such that

$$F(x) = 1 \iff \exists y \forall z \exists w M(x, y, z, w) = 1$$

   where all quantifiers are over strings of size polynomial in x. Show that for every constant c there exists a function F in Σ_3 which cannot be computed by circuits of size n^c. (Hint 1: same as above. Hint 2. y is a function $y : \{0,1\}^{(c+1)\log n} \to \{0,1\}$; $z = (z_1, z_2)$ where z_1 is a circuit, z_2 a function; w is a circuit.)